



Clinical Paper

An observational near-infrared spectroscopy study on cerebral autoregulation in post-cardiac arrest patients: Time to drop 'one-size-fits-all' hemodynamic targets?☆



K. Ameloot^{a,*,1}, C. Genbrugge^{b,c,1}, I. Meex^{b,c}, F. Jans^{b,c}, W. Boer^b, M. Vander Laenen^b, B. Ferdinande^a, W. Mullens^{a,c}, M. Dupont^a, J. Dens^{a,c}, C. DeDeyne^{b,c}

^a Department of Cardiology, Ziekenhuis Oost-Limburg, Genk, Belgium

^b Department of Anesthesiology and Critical Care Medicine, Ziekenhuis Oost-Limburg, Genk, Belgium

^c Faculty of Medicine and Life Sciences, University Hasselt, Diepenbeek, Belgium

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ABSTRACT

Aims: A subgroup of patients with ROSC after cardiac arrest (CA) with disturbed cerebral autoregulation might benefit from higher mean arterial pressures (MAP). We aimed to (1) phenotype patients with disturbed autoregulation, (2) investigate whether these patients have a worse prognosis, (3) define an individual optimal MAP per patient and (4) investigate whether time under this individual optimal MAP is associated with outcome.

Methods: Prospective observational study in 51 post-CA patients monitored with near infrared spectroscopy.

Results: (1) 18/51 patients (35%) had disturbed autoregulation. Phenotypically, a higher proportion of patients with disturbed autoregulation had pre-CA hypertension (31 ± 47 vs $65 \pm 49\%$, $p = 0.02$) suggesting that right shifting of autoregulation is caused by chronic adaptation of cerebral blood flow to higher blood pressures. (2) In multivariate analysis, patients with preserved autoregulation ($n = 33$, 65%) had a significant higher 180-days survival rate (OR 4.62, 95% CI [1.06:20.06], $p = 0.04$). Based on an index of autoregulation (COX), the average COX-predicted optimal MAP was 85 mmHg in patients with preserved and 100 mmHg in patients with disturbed autoregulation. (3) An individual optimal MAP could be determined in 33/51 patients. (4) The time under the individual optimal MAP was negatively associated with survival (OR 0.97, 95% CI [0.96:0.99], $p = 0.02$). The time under previously proposed fixed targets (65, 70, 75, 80 mmHg) was not associated with a differential survival rate.

Conclusion: Cerebral autoregulation showed to be disturbed in 35% of post-CA patients of which a majority had pre-CA hypertension. Disturbed cerebral autoregulation within the first 24 h after CA is associated with a worse outcome. In contrast to uniform MAP goals, the time spent under a patient tailored optimal MAP, based on an index of autoregulation, was negatively associated with survival.

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1. Introduction

In healthy individuals, cerebral autoregulation maintains cerebral perfusion across a broad range of changes in mean arterial pressure (MAP) with a presumed lower threshold between

50–60 mmHg.¹ Current post-cardiac arrest (CA) guidelines recommend targeting a MAP above 65 mmHg.² However, some post-CA patients might benefit from resuscitation to a higher MAP in order to preserve cerebral perfusion due to a rightward shift of cerebral autoregulation.³ Aiming for the same one-size-fits-all hemodynamic targets among all patients may potentially be harmful due to cerebral hypoperfusion in patients with a right-shifted autoregulation or due to excessive use of fluids and vasopressive agents in patients with preserved autoregulation. Moreover, experimental data revealed the possible influence of therapeutic hypothermia on the preservation of cerebral autoregulation, rendering it even more difficult to predict the optimal MAP ensuring adequate cerebral perfusion in individual patients.⁴ Post-CA patients have a

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* Corresponding author at: Ziekenhuis Oost-Limburg, Schiepse Bos, 3600 Genk, Belgium.

E-mail address: Koen.ameloot@zol.be (K. Ameloot).

¹ KA and CG equally contributed to this paper.

large cerebral penumbra at risk for secondary ischemic damage in case of suboptimal brain oxygenation. Brain tissue oxygen saturation (SctO₂) can be measured non-invasively with near-infrared spectroscopy.⁵ The present study was performed to investigate the relationship between MAP and SctO₂ in individual post-CA patients. Specific aims were (1) to investigate whether patients with preserved autoregulation have a better prognosis, (2) to compare patient characteristics of patients with preserved versus disturbed autoregulation, (3) to define an individual optimal MAP per patient and (4) to investigate whether the percentage of time under the predicted individual optimal MAP is associated with outcome.

2. Methods

2.1. Study population

All comatose survivors after non-traumatic CA treated in our tertiary care hospital (Ziekenhuis Oost-Limburg, Genk, Belgium) are prospectively enrolled in our database ($n = 82$). The present study includes all patients with continuous MAP and SctO₂ monitoring recorded every 2 s during the first 24 h of ICU stay between 2011 and 2013. Data from 31 patients, with recordings at a longer than 2 s time interval were excluded for the present analysis, leaving a study population of 51 patients. Patients could have been resuscitated in-hospital, referred by another hospital or admitted by our own emergency ward. All patients were treated uniformly according to the institutional post-CA protocol. As part of this protocol, patients are routinely monitored by cerebral saturation monitoring on hospital arrival. Written informed consent was obtained from a next of kin. Patients who died during the first 24 h were excluded from analysis. The study protocol was approved by the local medical ethics committee.

2.2. General management

Our institutional post-CA protocol has been described previously.^{5,6} In summary, all patients were intubated, mechanically ventilated and sedated with propofol and remifentanyl if hemodynamically tolerated. Cisatracurium was administered in case of shivering. Unless an obvious non-cardiac cause could be identified, all patients were referred for urgent coronary angiography followed by percutaneous coronary intervention when indicated. Therapeutic hypothermia was induced shortly after hospital admission by cold saline (4 °C – 30 ml/kg) and further mechanically induced and maintained in the coronary care unit by endovascular (Icy-catheter, CoolGard® 3000, Alsium, Irvine, CA, USA) or surface (ArcticGel™ pads, Arctic Sun® 5000, Medivance, Louisville, Colorado, USA) cooling systems at 33 °C for 24-h. After rewarming (0.3 °C/h) sedation was titrated toward patient's comfort with efforts toward minimizing sedation. Patients were extubated when their neurological, respiratory and hemodynamic status had recovered sufficiently.

2.3. Cerebral saturation monitoring

Cerebral tissue oxygen saturation was continuously measured with near infrared spectroscopy (NIRS), using the FORE-SIGHT™ technology (CAS Medical systems, Branford, CT, USA) and transmitted to a personal computer together with hemodynamic data with a 2 s time interval. Sensors were bilaterally applied to the frontotemporal area before the start of mechanically induced hypothermia. Cerebral saturation data were not used to guide any form of hemodynamic management.

2.4. Hemodynamic monitoring and management

Patients were treated according to the guidelines with the main focus on achieving a MAP above 65 mmHg.² If signs of inadequate circulation persisted despite correct fluid resuscitation (wedge pressure >18 mmHg), first norepinephrine was infused with a target MAP of 65 mmHg and subsequently dobutamine with a target cardiac index >2.2 l/min/m². An IABP was installed as deemed necessary by treating physicians ($n = 5$). A transthoracic echocardiography was performed in the first 24 h after admission. Blood pressures were obtained by radial artery line.

2.5. Statistics

Results are expressed as mean ($\pm$ SD, standard deviation) unless otherwise stated. The relationship between MAP and SctO₂ during the first 24 h after admission was investigated per patient in 3 steps (Fig. 1). First, for each patient separately the average SctO₂ was calculated per mmHg MAP (range 50–130 mmHg) to construct a MAP/SctO₂ scatterplot. Paired average SctO₂/MAP values were excluded if less than 10 paired measurements were available for a particular MAP out of a total of 43,200 pairs per patient per 24 h. Linear regression was performed and autoregulation was classified for each patient as preserved (slope of the linear regression equation <0.05%/mmHg or Pearson correlation coefficient not significant) or disturbed (slope of the linear regression equation >0.05%/mmHg and Pearson correlation coefficient significant). Patients with preserved and disturbed autoregulation were compared regarding their baseline characteristics by a student's *t*-test. To perform survival analysis, a multivariate logistic regression model was constructed including all significant univariate predictors of survival. Second, a MAP/COX bar graph was constructed per patient to predict the individual optimal MAP as previously described.⁴ Shortly, arterial pressure signals and NIRS data were digitalized and transferred to a laptop computer. COX was calculated as the moving linear correlation coefficient between 10 s averaged values of SctO₂ and MAP over moving 5 min time windows during the 24 h study period. In this way, COX is a continuous variable that ranges from –1 to +1 and can be used as a robust marker of autoregulation at a specific time point. Intact autoregulation is indicated by negative or near-zero COX because small MAP variations will not alter cerebral oxygenation when autoregulation is active. When autoregulation becomes impaired, COX becomes positive because MAP and SctO₂ correlate. We averaged all obtained COX values during the 24 h study period per 5 mmHg MAP bins. The optimal MAP as predicted by COX corresponds with the MAP bin with the most negative average COX value. In order to eliminate signal artifacts, we excluded bins compromising less than 1% of the 24 h recording period. Finally, the percentage of time under the COX predicted optimal MAP was calculated per patient and the association with outcome was evaluated by logistic regression. For comparison, the percentage of time under MAP 65, 70, 75 and 80 mmHg were calculated and evaluated the same way. Statistical analysis was performed using Matlab software (version R2010b, Mathworks, USA). A *p*-value <0.05 was considered significant.

3. Results

3.1. Study population

Fifty-one patients were included in the study. Baseline characteristics are summarized in Table 1. Twenty-five patients (49%) had a good outcome defined as survival in CPC 1–2 at 180 days post-CA.

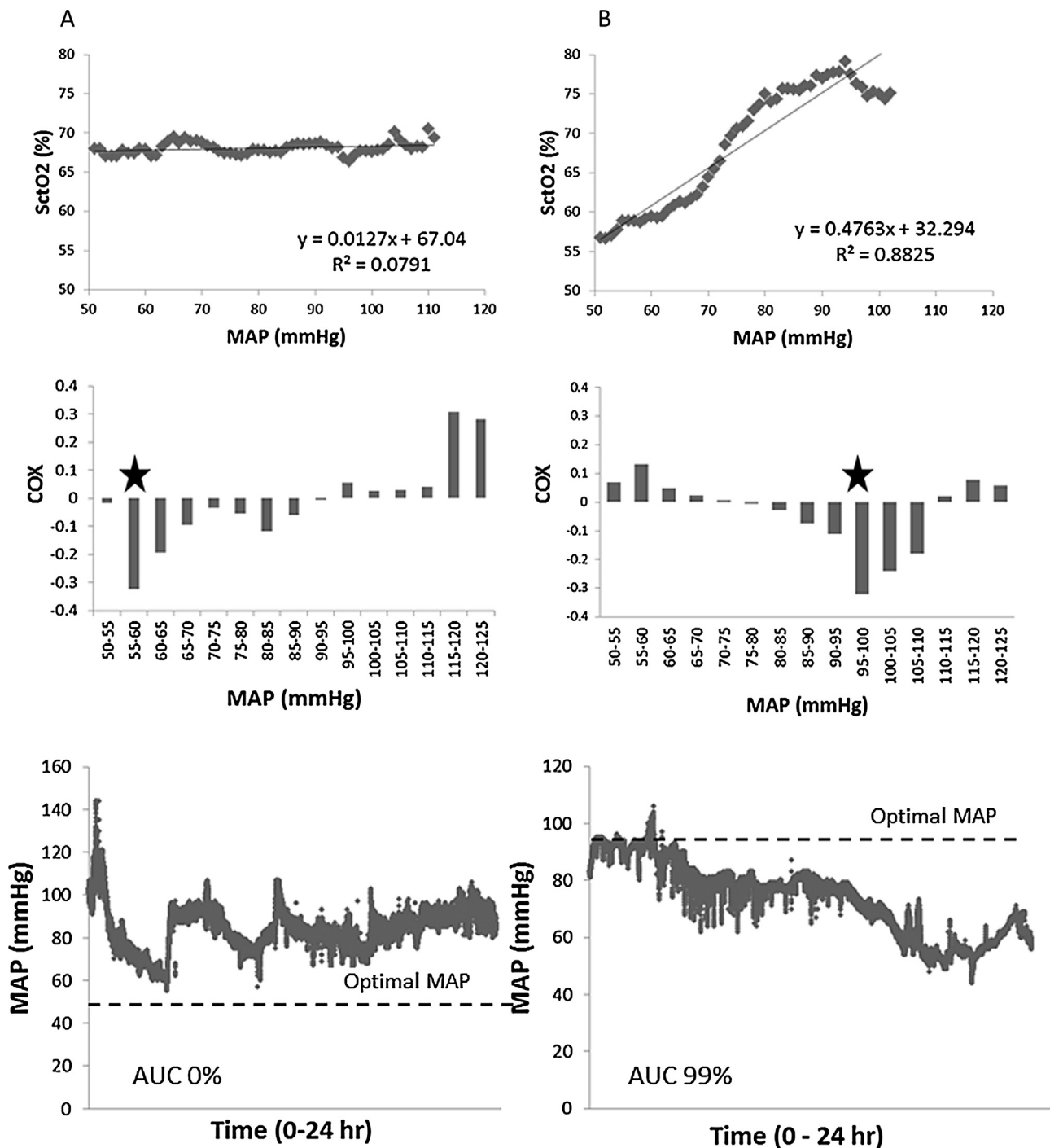


Fig. 1. Patient A shows preserved autoregulation with a COX predicted optimal MAP between 55 and 60 mmHg. During therapeutic hypothermia, the time under the optimal MAP was 0%. Patient was discharged alive 10 days after CA. Patient B shows right-shifted autoregulation with a COX based predicted optimal MAP between 95 and 100 mmHg. During therapeutic hypothermia, the time under the optimal MAP was 99%. Patient died at day 14 post-CA due to post-ischemic encephalopathy. * Cox predicted optimal MAP.

3.2. Cerebral autoregulation

Cerebral autoregulation was classified as disturbed in 18/51 (35%) of the patients. Chronic arterial hypertension was more frequent in patients with disturbed autoregulation (31 ± 47 vs $65 \pm 49\%$, $p = 0.02$). There were no other significant baseline differences between patients with preserved and disturbed autoregulation with respect to demographics, resuscitation characteristics, hemodynamic parameters, blood gasses, neuron specific

enolase (24h after admission) or cerebral saturation. In the subgroup with preserved autoregulation, 19/33 patients (57%) survived in CPC1-2 until 180 days post-CA compared with 6/18 (33%) of patients in the subgroup with disturbed. Significant predictors of good outcome in univariate analysis included the presence of bystander CPR ($p = 0.001$) and a presenting shockable rhythm ($p = 0.008$). Age ($p = 0.66$), left ventricular ejection fraction ($p = 0.08$) and admission lactate ($p = 0.45$) were not associated with survival in univariate analysis. When integrated in a multivariate logistic

Table 1
Comparison between patients with preserved and disturbed or right-shifted autoregulation.

	All	Cerebral autoregulation		
		Preserved	Disturbed right-shifted	p-Value
Number patients	51	33/51 (65%)	18/51 (35%)	
Demographics				
Age (years)	64 ± 13	63 ± 13	65 ± 13	0.70
Male (%)	62 ± 49	71 ± 47	53 ± 51	0.37
Chronic hypertension	43 ± 50	31 ± 47	65 ± 49	0.02
Resuscitation parameters				
Bystander CPR < 10 min (%)	80 ± 40	77 ± 43	86 ± 36	0.51
Duration ALS (min)	16 ± 13	14 ± 13	19 ± 14	0.24
Initial rhythm shockable (%)	69 ± 47	72 ± 46	63 ± 50	0.53
Hemodynamic parameters				
Heart rate (BPM)	65 ± 14	66 ± 14	64 ± 14	0.57
Echocardiographic LVEF (%)	41 ± 18	40 ± 19	42 ± 16	0.82
Mean MAP/24 h (mmHg)	77 ± 8	76 ± 8	78 ± 8	0.35
Mean SVO ₂ /24 h (%)	68 ± 10	68 ± 9	68 ± 12	0.95
Mean CO/24 h (l/min)	3.70 ± 1.26	3.87 ± 1.25	3.36 ± 1.25	0.24
Mean cerebral saturation/24 h (%)	66 ± 5	66 ± 4	66 ± 5	0.93
Blood gas				
First lactate (mmol/l)	6 ± 5	6 ± 5	7 ± 5	0.35
Mean lactate/24 h (mmol/l)	3 ± 2	3 ± 2	3 ± 1	0.65
Mean pH/24 h	7.33 ± 0.07	7.32 ± 0.07	7.34 ± 0.06	0.36
Mean pCO ₂ /24 h (mmHg)	41 ± 5	41 ± 5	41 ± 4	0.75
Mean pO ₂ /24 h (mmHg)	103 ± 23	102 ± 26	103 ± 16	0.88
Neuron specific enolase/24 h	69 ± 61	64 ± 66	76 ± 50	0.50
Survival (%)	25/51 (49%)	19/33 (57%)	6/18 (33%)	0.03

regression model, a preserved autoregulation persisted as an independent predictor of good outcome (OR 4.62, 95% CI [1.06;20.06], $p=0.04$) (Table 2).

3.3. COX predicted optimal MAP

When COX per 5 mmHg bar graphs were pooled together thereby averaging all 51 study patients, the overall COX predicted optimal MAP was between 85 and 105 mmHg. When a pooled analysis was performed for all 18 patients with disturbed autoregulation, the overall COX predicted optimal MAP was 100 mmHg. When a pooled analysis was performed for all 33 patients with preserved autoregulation, the overall COX predicted optimal MAP was 85 mmHg (Fig. 2).

Based on average COX per 5 mmHg MAP bar graphs, an optimal MAP could be identified in 33/51 (65%) of the patients. Fourteen of these patients (42%) had a favorable outcome. The COX predicted optimal MAP was plotted per patient against the median MAP during the 24 h study period (Fig. 3). The time under the COX predicted optimal MAP was negatively associated with survival (OR 0.97, 95% CI [0.96;0.99] per % of time, $p=0.02$). The time under other suggested fixed targets (65, 70, 75, 80 mmHg) was not associated with a differential survival rate (Table 3).

4. Discussion

First, we showed that cerebrovascular autoregulation is disturbed in approximately one third of post-CA patients during

therapeutic hypothermia. Presumably, in most patients, we should interpret the disturbed autoregulation as a right-shift of its lower limits. Two-thirds of these patients had chronic arterial hypertension pre-cardiac arrest suggesting that the right shifting of the lower threshold of autoregulation may be caused by protective chronic adaption of the cerebral blood flow to elevated blood pressures. Importantly, these patients had a poorer prognosis when compared with patients with preserved autoregulation. Subgroup analysis did not show evidence for more severe baseline brain damage in the subgroup with disturbed or right-shifted autoregulation since rates of basic life support, presence of shockable rhythm, time-to-ROSC, initial lactate and levels of neuron specific enolase after 24 h were all comparable in both subgroups. In addition, all ICU parameters including global hemodynamics and blood gasses were also comparable in both subgroups. Moreover, impaired autoregulation was statistically independently associated with an unfavorable outcome, suggesting that impaired autoregulation per se might be an important pathophysiological factor that causes additional brain damage after ROSC. All these observations support the hypothesis that patients with right-shifted autoregulation benefit from higher MAP's to ensure cerebral perfusion. Alternatively, since we did not measure intracranial pressure (ICP) invasively, we cannot exclude the possibility that both the right-shifting of cerebral autoregulation and the observed worse outcome are due to more cerebral edema and higher ICP's in this subgroup.⁷ This is the first study to show an association between disturbed (or right-shifted) autoregulation and poor survival rates in adult post-CA patients in the first 24 h post-CA during

Table 2
Multivariate analysis.

	Odds ratio	Lower limit	Upper limit	p-Value
Initial shockable rhythm (yes/no)	9.38	1.6	54.78	0.01
Presence of basic life support (yes/no)	10.88	1.02	115.71	0.04
Preserved autoregulation (yes/no)	4.62	1.06	20.06	0.04

Table 3

The percentage of time under several fixed MAP targets is compared with the percentage of time under the COX predicted optimal MAP.

	Odds ratio	Lower limit	Upper limit	p-Value
% time < 65 mmHg	0.97	0.95	1.01	0.27
% time < 70 mmHg	0.98	0.96	1.00	0.13
% time < 75 mmHg	0.98	0.96	1.00	0.19
% time < 80 mmHg	0.99	0.97	1.01	0.46
% time < optimal MAP	0.97	0.96	0.99	0.02

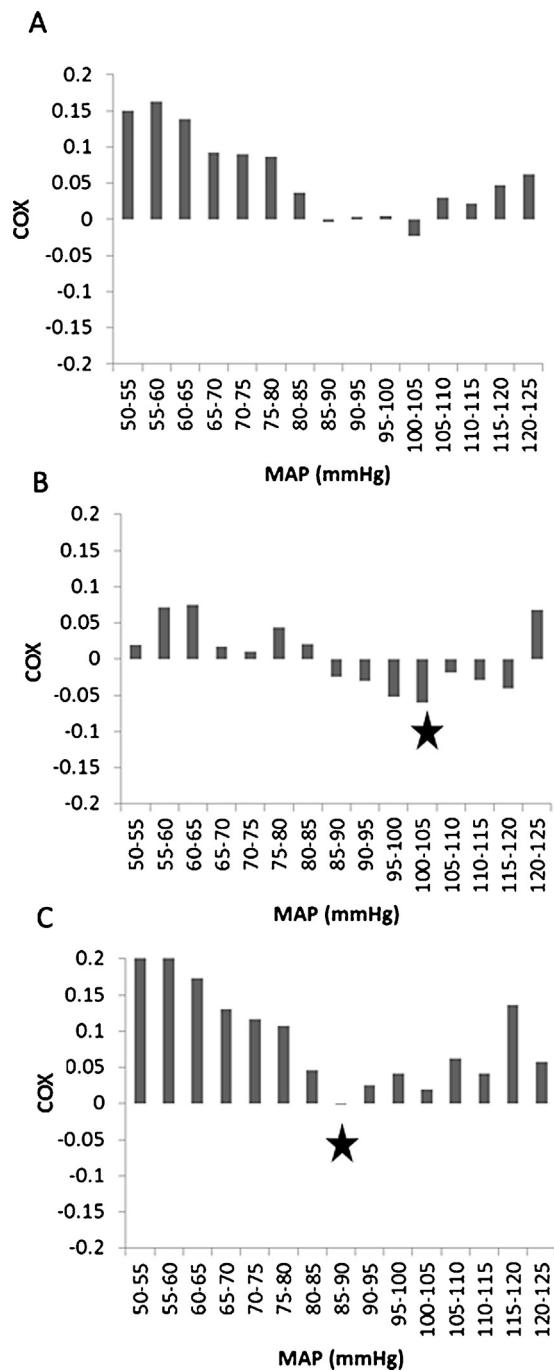


Fig. 2. Pooled COX per 5 mmHg MAP plot. (A) For all 51 study patients, (B) for 18 patients with disturbed/right shifted autoregulation, (C) for 33 patients with preserved autoregulation. * Cox predicted optimal MAP.

treatment with hypothermia (33°C). In line with our results, disturbed autoregulation has been shown to be associated with unfavorable outcome after traumatic brain injury,⁸ subarachnoid hemorrhage⁹ and during cardiopulmonary bypass.¹⁰

Our results are in line with a previous study with transcranial Doppler in normothermic post-CA patients showing a right-ward shift of cerebral autoregulation in a subset of post-CA patients.³ In contrast, experimental data from Lee et al. in an experimental model of pediatric cardiac arrest treated with hypothermia revealed a left-shift of the lower limit of autoregulation, particularly associated to the use of hypothermia.⁴ Therefore, cerebral autoregulatory status during therapeutic hypothermia in a post-CA setting is still unclear. Bouzat et al. used norepinephrine to

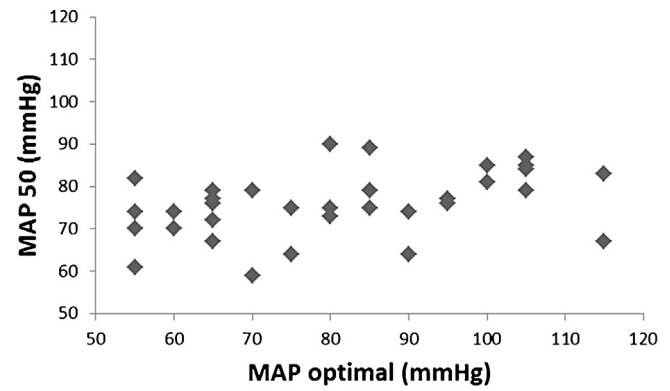


Fig. 3. The relationship between COX predicted optimal MAP and median MAP during the 24 h study period.

increase MAP from 72 to 90 mmHg in 10 post-CA patients during the maintenance phase of therapeutic hypothermia.¹¹ They did not observe a significant increase in SctO₂ (from 73.6 to 74.1%) and concluded that dynamic cerebrovascular autoregulation was preserved in hypothermic patients in the MAP range studied.

Second, we found the average COX predicted optimal MAP in post-CA patients to be between 85 and 105 mmHg. Therefore it seems that a MAP of 65 mmHg is insufficient in most post-CA patients, regardless of the autoregulatory status. In the subgroup of patients with disturbed (right-shifted) autoregulation, the overall COX predicted optimal MAP was 100 mmHg compared with 85 mmHg in the subgroup with preserved autoregulation. In only 6 patients, the COX predicted optimal MAP was equal or smaller than the guideline recommended MAP of 65 mmHg. Our findings are in agreement with 2 retrospective and 1 prospective studies previously published on the positive association between higher MAP's during the first hour's post-ROSC and increased survival in post-CA patients.^{12–14}

Finally, the percentage of time under the COX predicted optimal MAP showed to be a far better predictor of mortality than the percentage of time under any other fixed MAP target. In recent years, multiple NIRS-derived indices of autoregulation with different devices were validated against transcranial Doppler in animal models,^{15,16,4,17} during cardiopulmonary bypass^{10,18,19} and in patients with severe sepsis²⁰ and subarachnoid hemorrhage.⁹ Some of these indices have been dynamically validated against laser-Doppler flowmetry in animal models with balloon induced blood pressure changes.^{4,17} In this way, COX should be regarded as a static index of autoregulation that indicates whether autoregulation is (in)active at a certain time point with a given MAP, cardiac output and carbon dioxide level. Most COX validation studies were performed with the INVOS device that can be used interchangeably with the Foresight device that was used in the present study.²¹ Our results are in line with two recent observational studies showing that deviation from the blood pressure with optimal autoregulatory vasoreactivity predicts poor neurologic outcomes and is associated with more cerebral damage on MRI after pediatric or neonatal cardiac arrest.^{22,23} This study is unique since we showed for the first time an association between outcome and patient tailored hemodynamic goals based on thresholds of cerebrovascular autoregulation as assessed by NIRS in adult post-CA patients.

Our findings have at least 2 potential important consequences for daily clinical practice. First, based on our results, we should target MAP's above the currently recommended 65 mmHg to ensure optimal cerebral perfusion within the autoregulatory zone in most post-CA patients and especially in the subgroup of patients with chronic hypertension and right-shifted autoregulation. Second,

our results suggest that we should attempt to individualize our hemodynamic goals. The added value of continuous cerebral perfusion monitoring (by SctO₂ or by transcranial Doppler) becomes clear in the light of the very large variations of the optimal MAP among different patients, of SctO₂ at a given MAP between different patients and between different time points in the same patient. Perhaps, in analogy with continuous stroke volume and pulse pressure variation to predict fluid responsiveness, minute-to-minute COX per MAP monitors can be developed to continuously predict cerebral responsiveness to blood pressure changes. However, only an interventional trial can determine whether forcing MAP's closer to the COX predicted optimal MAP with additional fluids and pharmacological support is feasible and can improve outcome.

A first limitation is the rather small sample size. Since a COX predicted optimal MAP could only be determined in 33 patients, multivariate analysis could not be performed due to the risk of overfitting. In patients with preserved autoregulation, SctO₂ remains stable among a large MAP range and COX values per 5 mmHg MAP bins are therefore often equally negative. In some patients with extreme right-shifting of autoregulation the optimal MAP is never reached and COX values per 5 mmHg are all positive. Therefore, given the small sample size the results of this pilot trial should be regarded as hypothesis generating and be confirmed in a larger validation study. Although some might consider it a limitation that we excluded 5 patients with refractory cardiogenic shock who died during the first 24 h of the study, we believe that post-CA guidelines should be designed for patients with the largest potential benefit of a goal directed neuroprotective hemodynamic optimization strategy. Obviously, it is not possible to force MAP in a higher range using additional fluids or vasopressive agents in a patient already receiving maximal pharmacological support. Second, the COX predicted optimal MAP as presented here represents an average value over the 24-hours study period. We cannot exclude the possibility that the optimal MAP changes dynamically during induction and maintenance of hypothermia. A third limitation is the absence of detailed information on vasopressor and sedation use during the study period. It cannot be excluded that use of norepinephrine changes the lower threshold of cerebral autoregulation. However, treating physicians were all instructed to maintain a MAP of 65 mmHg and a cardiac index of at least 2.2 l/min/m² with the lowest amount of pharmacological support. Fourth, we used NIRS as surrogate measure for cerebral perfusion since this is easy to use and non-invasive. Since SctO₂ reflects the balance between cerebral oxygen supply and consumption, some authors prefer to use transcranial Doppler as a means of assessing cerebral perfusion. Near-infrared spectroscopy was previously validated against jugular bulb saturation in healthy volunteers and against CT-perfusion for the assessment of regional cerebral perfusion in patients with traumatic brain injury.^{21,24}

In summary, cerebral autoregulation is disturbed, probably right-shifted in about one third of post-CA patients of which a majority had pre-CA hypertension. These patients have a worse prognosis as they are at risk for cerebral hypoperfusion when resuscitated to uniform hemodynamic targets and have worse prognosis. In contrast to these uniform MAP goals, the time under a patient tailored optimal MAP, based on an index of autoregulation, was negatively associated with survival. However, whether maintaining MAP's closer to the COX predicted optimal MAP would improve outcome cannot yet be determined.

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Conflict of interest statement

None declared.

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References

- Drummond JC. The lower limit of autoregulation: time to revise our thinking? *Anesthesiology* 1997;86:1431–3.
- Peberdy MA, Callaway CW, Neumar RW, et al. American Heart Association. Post-cardiac arrest care: 2010 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation* 2010;122:S768–86.
- Sundgreen C, Larsen FS, Herzog TM, Knudsen GM, Boesgaard S, Aldershvile J. Autoregulation of cerebral blood flow in patients resuscitated from cardiac arrest. *Stroke* 2001;32:128–32.
- Lee JK, Yang ZJ, Wang B, et al. Noninvasive autoregulation monitoring in a swine model of pediatric cardiac arrest. *Anesth Analg* 2012;114:825–36.
- Meex I, Dens J, Jans F, et al. Cerebral tissue oxygen saturation during therapeutic hypothermia in post-cardiac arrest patients. *Resuscitation* 2013;84:788–93.
- Ameloot K, Meex I, Genbrugge C, et al. Accuracy of continuous thermodilution cardiac output monitoring by pulmonary artery catheter during therapeutic hypothermia in post-cardiac arrest patients. *Resuscitation* 2014;85:1263–8.
- Brady KM, Lee JK, Kibler KK, et al. The lower limit of cerebral blood flow autoregulation is increased with elevated intracranial pressure. *Anesth Analg* 2009;108:1278–83.
- Czosnyka M, Smielewski P, Kirkpatrick P, Menon DK, Pickard JD. Monitoring of cerebral autoregulation in head-injured patients. *Stroke* 1996;27:1829–34.
- Soehle M, Czosnyka M, Pickard JD, Kirkpatrick PJ. Continuous assessment of cerebral autoregulation in subarachnoid hemorrhage. *Anesth Analg* 2004;98:1133–9.
- Ono M, Joshi B, Brady K, et al. Risks for impaired cerebral autoregulation during cardiopulmonary bypass and postoperative stroke. *Br J Anaesth* 2012;109:391–8.
- Bouzat P, Suys T, Sala N, Oddo M. Effect of moderate hyperventilation and induced hypertension on cerebral tissue oxygenation after cardiac arrest and therapeutic hypothermia. *Resuscitation* 2013;84:1540–5.
- Beylin ME, Perman SM, Abella BS, et al. Higher mean arterial pressure with or without vasoactive agents is associated with increased survival and better neurological outcomes in comatose survivors of cardiac arrest. *Intensive Care Med* 2013;39:1981–8.
- Trzeciak S, Jones AE, Kilgannon JH, et al. Significance of arterial hypotension after resuscitation from cardiac arrest. *Crit Care Med* 2009;37:2895–903.
- Kilgannon JH, Roberts BW, Jones AE, et al. Arterial blood pressure and neurological outcome after resuscitation from cardiac arrest. *Crit Care Med* 2014;42:2083–91.
- Brady KM, Lee JK, Kibler KK, et al. Continuous time-domain analysis of cerebrovascular autoregulation using near-infrared spectroscopy. *Stroke* 2007;38:2818–25.
- Lee JK, Kibler KK, Benni PB, et al. Cerebrovascular reactivity measured by near-infrared spectroscopy. *Stroke* 2009;40:1820–6.
- Lee JK, Brady KM, Mytar JO, et al. Cerebral blood flow and cerebrovascular autoregulation in a swine model of pediatric cardiac arrest and hypothermia. *Crit Care Med* 2011;39:2337–45.
- Joshi B, Ono M, Brown C, et al. Predicting the limits of cerebral autoregulation during cardiopulmonary bypass. *Anesth Analg* 2012;114:503–10.
- Brady K, Joshi B, Zweifel C, et al. Real-time continuous monitoring of cerebral blood flow autoregulation using near-infrared spectroscopy in patients undergoing cardiopulmonary bypass. *Stroke* 2010;41:1951–6.
- Steiner LA, Pfister D, Strebel SP, Radolovich D, Smielewski P, Czosnyka M. Near-infrared spectroscopy can monitor dynamic cerebral autoregulation in adults. *Neurocrit Care* 2009;10:122–8.
- MacLeod DB, Ikeda K, Vacchiano P. Absolute and trending accuracy of forehead and INVOS cerebral oxymeters in healthy volunteers. In: Proceedings of the annual meeting of the American society of Anesthesiologists. 2009. p. A298.
- Lee JK, Brady KM, Chung SE, et al. A pilot study of cerebrovascular reactivity autoregulation after pediatric cardiac arrest. *Resuscitation* 2014;85:1387–93.
- Tekes A, Poretti A, Scheurkogel MM, et al. Apparent diffusion coefficient scalars correlate with near-infrared spectroscopy markers of cerebrovascular autoregulation in neonates cooled for perinatal hypoxic-ischemic injury. *AJNR Am J Neuroradiol* 2014 <http://www.ajnr.org/content/early/2014/08/28/ajnr.A4083.long>
- Taussky P, O'Neal B, Daugherty WP, et al. Validation of frontal near-infrared spectroscopy as noninvasive bedside monitoring for regional cerebral blood flow in brain-injured patients. *Neurosurg Focus* 2012;32.